(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(CBCS)
Course Name: B.Tech. (MAE), Semester: 3rd
(Oct, 2022)

Subject Code: BMA205 **Subject: Thermal Engineering - I**
Time : 1 ½ Hours
Note: Q. 1 is compulsory. Attempt any two questions from the rest. **Maximum Marks : 30**

Q1 **(2.5*4)**

(a) If $pv=RT$, determine whether $\int\left(\frac{dT}{T}+\frac{pdv}{v}\right)$ can be used as property or not.

(b) Define a throttle process and apply steady flow energy equation to it.

(c) Draw the constant volume heat addition process between the states (T_1, s_1) and (T_2, s_2) and the constant pressure heat addition process between the states (T_1, s_1) and (T_2, s_2) on the same T-s plane.

(d) Write the four Maxwell equations.

Q2 **(5,5)**

(a) Derive the expression for the heat transfer $({}_1q_2)$ during a polytropic process 1-2 in terms of the work done $({}_1w_2)$, specific heat ratio (γ) and polytropic index (n).

(b) In the condenser of a steam power plant, the steam flows steadily at the rate of 300 kg/hr. At inlet to the condenser, the steam has an enthalpy of 2050 kJ/kg and its velocity is 500 m/s. At outlet of the condenser, the condensate (condensed steam) has an enthalpy of 200 kJ/kg and it leaves with 10 m/s velocity. The outlet is in line with the inlet. Determine the heat transfer from the steam to the cooling water in kJ/s.

Q3 **(5,5)**

(a) Derive the expression for the entropy change during a polytropic process in terms of C_p, R, temperature ratio and pressure ratio.

(b) An engineer claims to have built an engine that absorbs 330 kJ of heat from a hot reservoir at 370 K and delivers 100 kJ of work. He further states that his engine rejects heat to two cold reservoirs: 130 kJ to a reservoir at 300 K and 100 kJ to another reservoir at 250K. Apply both the first and the second laws of thermodynamics to check whether his claim is valid or not.

Q4 **(5,5)**

(a) A rigid container containing 0.005 m^3 of nitrogen gas at 100 kPa and 290 K is heated reversibly until the temperature becomes 360 K. Calculate the heat supplied in kJ and the entropy change in kJ/K. Assume nitrogen to be a perfect gas and for this gas, take R = 196.93 J/kg K and $\gamma = 1.4$.

(b) Derive the expression for $\frac{C_p}{C_v}$ in terms of $\left(\frac{\partial p}{\partial v}\right)_s$ and $\left(\frac{\partial p}{\partial v}\right)_T$.